

1. MC.08.01

Which of the following represents a step in the short-run decision-making model?

- ☐ a. Defining the problem
- ☐ b. Identifying alternatives
- ☐ c. Identifying the costs and benefits of feasible alternatives
- ☐ d. Assessing qualitative factors
- ☒ e. All of these choices are steps in the short-run decision-making model.



Feedback

Check My Work

The steps for short-run decision-making are: recognize and define problem, identify alternatives, identify costs and benefits, estimate relevant costs and benefits, assess qualitative factors and make decision.

Solution

e

2. MC.08.02.ALGO (Algorithmic)

Costs that *cannot* be affected by any future action are called

- ☐ a. differential costs.
- ☐ b. product costs.
- ☐ c. relevant costs.
- ☒ d. sunk costs.
- ☐ e. joint costs.



Feedback

Check My Work

Relevant costs are future costs and differ across alternatives. A sunk cost can not be affected by any future action. Joint costs are common costs shared in production before split-off.

Solution

d

3. MC.08.03.ALGO (Algorithmic)

Sandy is considering moving from her apartment into a small house with a fenced yard. The apartment is noisy, and she has difficulty studying. In addition, the fenced yard would be great for her dog. The distance from school is about the same from the house and from the apartment. The apartment costs \$750 per month, and she has 2 months remaining on her lease. The lease cannot be broken, so Sandy must pay the last 2 months of rent whether she lives there or not. The rent for the house is \$450 per month, plus utilities, which should average \$100 per month. The apartment is furnished; the house is not. If Sandy moves into the house, she will need to buy a bed, dresser, desk, and chair immediately. She thinks that she can pick up some used furniture for a good price.

Which of the following costs is *irrelevant* to Sandy's decision to stay in the apartment or move to the house?

- ☐ a. The cost of the used furniture
- ☒ b. The last 2 months of rent in the apartment
- ☐ c. The cost of the dresser
- ☐ d. Utilities for the house of \$100 per month
- ☐ e. House rent of \$450 per month



Feedback

Check My Work

Irrelevant costs occur when future costs are the same for more than one alternative. Then they have no effect on the decision.

Solution

4. MC.08.04.ALGO (Algorithmic)

Sandy is considering moving from her apartment into a small house with a fenced yard. The apartment is noisy, and she has difficulty studying. In addition, the fenced yard would be great for her dog. The distance from school is about the same from the house and from the apartment. The apartment costs \$750 per month, and she has 2 months remaining on her lease. The lease cannot be broken, so Sandy must pay the last 2 months of rent whether she lives there or not. The rent for the house is \$450 per month, plus utilities, which should average \$100 per month. The apartment is furnished; the house is not. If Sandy moves into the house, she will need to buy a bed, dresser, desk, and chair immediately. She thinks that she can pick up some used furniture for a good price.

Which of the following is a qualitative factor?



- ☐ a. Apartment rent of \$750 per month
- ☒ b. The noise in the apartment house
- ☐ c. Utilities for the house of \$100 per month
- ☐ d. The cost of food for her dog
- ☐ e. The cost of the used furniture

Feedback

Check My Work

Qualitative factors are difficult to associate an amount with but include issues such as political pressure and product safety.

Solution

b

5. MC.08.05.ALGO (Algorithmic)

Sandy is considering moving from her apartment into a small house with a fenced yard. The apartment is noisy, and she has difficulty studying. In addition, the fenced yard would be great for her dog. The distance from school is about the same from the house and from the apartment. The apartment costs \$750 per month, and she has 2 months remaining on her lease. The lease cannot be broken, so Sandy must pay the last 2 months of rent whether she lives there or not. The rent for the house is \$450 per month, plus utilities, which should average \$100 per month. The apartment is furnished; the house is not. If Sandy moves into the house, she will need to buy a bed, dresser, desk, and chair immediately. She thinks that she can pick up some used furniture for a good price.

Suppose that the apartment building was within walking distance to campus and the house was five miles away. Sandy does not own a car. How would that affect her decision?



- ☐ a. It would make both choices more desirable.
- ☒ b. It would make the apartment more desirable.
- ☐ c. It would make both choices equally more undesirable.
- ☐ d. It would make her consider buying a bicycle instead of a car.
- ☐ e. It would make the house more desirable.

Feedback

Check My Work

Consider the options and best situation for Sandy.

Solution

b

6. MC.08.06

Which of the following items could represent a qualitative factor in relevant decision-making for a clothing manufacturer?



- ☐ a. The amount of Scope 1 and Scope 2 GHG emissions occurring in production.
- ☐ b. The percentage of recycled materials used in packaging finished goods.
- ☐ c. The number of gallons of water used during production.
- ☒ d. All of the above.

☐ e. None of the above.

Feedback

Check My Work

Correct

Solution

d

7. MC.08.07

Which of the following statements is false?

- ☐ a. Fixed costs are never relevant.
- ☐ b. Variable costs are never relevant.
- ☐ c. Usually, variable costs are irrelevant.
- ☐ d. Step costs are irrelevant when a decision alternative requires moving outside of the existing relevant range.
- ✓ ☒ e. All of these choices are correct.

Feedback

Check My Work

Relevant costs are future costs and differ across alternatives. Irrelevant costs occur when future costs are the same for more than one alternative. Then they have no effect on the decision.

Solution

e

8. MC.08.08

In a segmented income statement, which of the following statements is true?

- ☐ a. Segment margin is greater than contribution margin.
- ☐ b. Common fixed expenses must be allocated to each segment.
- ☐ c. Contribution margin is equal to sales less all variable and direct fixed expenses of a segment.
- ☐ d. Segment margin is equal to contribution margin less direct and common fixed expenses.
- ✓ ☒ e. Segment margin is equal to contribution margin less direct fixed expenses.

Feedback

Check My Work

Segmented income statements allow managers to see the profitability of individual segments of the company, which is helpful when making add-or-drop decisions.

Solution

e

9. MC.08.09.ALGO (Algorithmic)

In a make-or-buy decision,

- ☐ a. the company must choose between raising or lowering the average selling price on its products.
- ☐ b. the company must choose whether or not to close a business unit.
- ✓ ☒ c. the company would consider the purchase price of the externally provided good to be relevant.
- ☐ d. the company would consider all fixed overhead to be relevant.
- ☐ e. None of these choices are correct.

Feedback

Check My Work

Make-or-buy decisions involve choices between internal and external production.

Solution

10. MC.08.10

Using a complex mathematical algorithm to help identify the optimal make-or-buy decision outcome represents an example of [Refer to [Exhibit 2-2](#)]:

- ✓
- ☐ a. Diagnostic data analytics
 - ☒ b. Prescriptive data analytics
 - ☐ c. Descriptive data analytics
 - ☐ d. Predictive data analytics
 - ☐ e. All of the above

Feedback

Check My Work
Correct
Solution
b

11. MC.08.11

Carroll Company, a manufacturer of vitamins and minerals, has been asked by a large drugstore chain to provide bottles of vitamin E. The bottles would be labeled with the name of the drugstore chain, and the chain would pay Carroll \$2.30 per bottle rather than the \$3.00 regular price. Which type of a decision is this?

- ✓
- ☐ a. Make-or-buy
 - ☒ b. Special-order
 - ☐ c. Keep-or-drop
 - ☐ d. Economic order quantity
 - ☐ e. Markup pricing

Feedback

Check My Work

Special-order decisions focus on accepting or rejecting a specially priced order. These orders are normally attractive, allowing the opportunity to utilize maximum productive capacity and profits. Make-or-buy decisions involve choices between internal and external production.

Solution
b

12. MC.08.12

A segment could be which of the following?

- ✓
- ☐ a. Product
 - ☐ b. Customer type
 - ☐ c. Geographic region
 - ☒ d. All of these.
 - ☐ e. None of these.

Feedback

Check My Work

A segment is a subunit of a company of sufficient importance to warrant the production of performance reports.

Solution
d

13. MC.08.13

Garrett Company provided the following information:

Product 1 Product 2

Units sold	10,000	20,000
Price	\$20	\$15
Variable cost per unit	\$10	\$10
Direct fixed cost	\$35,000	\$75,000

Common fixed cost totaled \$46,000. Garrett allocates common fixed cost to Product 1 and Product 2 on the basis of sales. If Product 2 is dropped, which of the following is true?

- 
- ☐

a. Sales will increase by \$300,000.
- ☐

b. Overall operating income will increase by \$2,600.
- ☒

c. Overall operating income will decrease by \$25,000.
- ☐

d. Overall operating income will not change.
- ☐

e. Common fixed cost will decrease by \$27,600.

Feedback

Check My Work

Prepare a segmented income statement for Product 2 to determine the amount of operating income the company will no longer earn if the product is dropped.

The most important aspect of this analysis is determining which costs will be eliminated (i.e., be relevant) and which costs will remain (i.e., be irrelevant) regardless of whether or not the segment or product is dropped.

Solution
c

14. MC.08.14

Jennings Hardware Store marks up its merchandise by 30%. If a part costs \$25.00, which of the following is true?

- 
- ☐

a. The price is \$7.50.
- ☐

b. The markup is \$32.50.
- ☒

c. The price is \$32.50.
- ☐

d. The markup is pure profit.
- ☐

e. All of these choices are correct.

Feedback

Check My Work

Price With Markup = Cost per Unit + (Cost per Unit × Markup Percentage)

Solution
c

15. MC.08.15

When a company faces a production constraint or scarce resource (e.g., only a certain number of machine hours are available), it is important to

- 
- ☐

a. produce the product with the highest contribution margin in total.
- ☐

b. produce the product with the lowest full manufacturing cost.
- ☒

c. produce the product with the highest contribution margin per unit of scarce resource.
- ☐

d. produce the product with the highest contribution margin per unit.
- ☐

e. The constraint is not relevant to the production problem.

Feedback

Check My Work

Unit Contribution Margin of Scarce Resource = Unit Selling Price – Unit Variable Cost/Required Amount of Scarce Resource per Unit

Solution
c

16. MC.08.16

In the keep-or-drop decision, the company will find which of the following income statement formats most useful?

✓

a. A segmented income statement in the contribution margin format

b. A segmented income statement in the full costing format that is used for financial reporting

c. An overall income statement in the contribution margin format

d. An overall income statement in the full costing format that is used for financial reporting

e. Income statements are of no use in making this type of decision.

Feedback

Check My Work

Segmented reports prepared on a variable-costing basis provide valuable information for keep-or-drop decisions.

Solution

a

17. MC.08.17

In the sell-or-process-further decision,

✓

a. joint costs are always relevant.

b. total costs of joint processing and further processing are relevant.

c. all costs incurred prior to the split-off point are relevant.

d. the most profitable outcome may be to further process some separately identifiable products beyond the split-off point beyond the split-off point.

e. None of these choices are correct.

Feedback

Check My Work

At times it is more profitable to process a joint product further, beyond the split-off point.

Solution

d

18. MC.08.45

A Plus Distributing Company is considering a special order for 1,000 units to be priced at \$9.10 (the normal price would be \$11.50). The order would require specialized materials costing \$4.00 per unit. Direct labor and variable factory overhead would cost \$2.15 per unit. Fixed factory overhead is \$1.20 per unit. However, the company has excess capacity and acceptance of the order would not raise total fixed factory overhead. The warehouse, however, would have to add capacity costing \$1,200. Which of the following items is relevant to the special order?

✓

a. Normal selling price of \$11.50

b. Fixed factory overhead per unit of \$1.20

c. Selling price per unit of special order of \$9.10

d. \$7.35 spent on donuts and coffee

Feedback

Check My Work

Correct

Post-Submission

The selling price of \$9.10 per unit is relevant to the special order for 1,000 units.

19. MC.08.62

The operations of Smits Corporation are divided into the Child division and the Jackson division. Projections for the next year are as follows:

	Child Division	Jackson Division	Total
Sales revenue	\$250,000	\$180,000	\$430,000

Variable expenses	90,000	100,000	190,000
Contribution margin	\$160,000	\$80,000	\$240,000
Direct fixed expenses	85,000	62,500	147,500
Segment margin	\$75,000	\$17,500	\$92,500
Allocated common costs	35,000	27,500	62,500
Total relevant benefit (loss)	\$40,000	\$(10,000)	\$30,000

Operating income for Smits Corporation as a whole, if the Jackson division were dropped, would be:

- ✓ ☒ a. \$12,500.
- ☐ b. \$10,500.
- ☐ c. \$10,000.
- ☐ d. \$15,000.

Feedback

Check My Work
Correct
Post-Submission

SUPPORTING CALCULATIONS:

$$\$75,000 - \$62,500 = \$12,500$$

20. MC.08.68

Tampa International Company manufactures a product with the following costs per unit at the expected production of 40,000 units:

Direct materials	\$5
Direct labor	10
Variable overhead	7
Fixed overhead	9

The company has the capacity to produce 50,000 units. The product regularly sells for \$50. A wholesaler has offered to pay \$42 per unit for 3,000 units.

If the firm chooses to accept the special order and reject some regular sales, the effect on operating income would be a(n):

- ✓ ☒ a. increase of \$30,000.
- ☐ b. decrease of \$24,000.
- ☐ c. decrease of \$21,000.
- ☐ d. increase of \$64,000.

Feedback

Check My Work
Correct
Post-Submission

$$\text{Effect on Operating Income} = 3,000 \text{ units} \times (\$50 - \$42) = \$24,000 \text{ decrease}$$

21. MC.08.83

Peach Company makes telephones. Currently, Peach makes all the components of the telephones in-house. An outside company has offered to supply one component, Part X76, for \$15 each. Peach uses 17,500 of these components per year. Costs of X76 are as follows:

Direct materials	\$2
Direct labor	2
Variable overhead	3
Fixed overhead	6

Assume that all of the fixed overhead is allocated and cannot be avoided. Should Peach purchase the part from the outside supplier?

- ☐ a. No, income will decrease by \$120,000.
- ☐ b. Yes, income will increase by \$140,000.
- ☐ c. Yes, income will increase by \$120,000.
- ☒ d. No, income will decrease by \$140,000.



Feedback

Check My Work
Correct
Post-Submission

Relevant Cost per Unit = Direct Materials + Direct Labor + Variable Overhead = \$2 + \$2 + \$3 = \$7
Decrease in Income if Purchased = (\$7 – \$15) × 17,500 units = \$(140,000)
Income will decrease by \$140,000 if the part is purchased.

22. MC.08.90

Consider the following portion of a segmented income statement for the year just ended. Assume fixed expenses of Division X include \$30,000 of direct expenses and that the discontinuance of the department will neither affect the sales of the other departments nor reduce the common expenses.

	Division X
Sales	\$110,000
Variable costs	60,000
Gross profit	\$50,000
Fixed expenses (direct and selling and administrative)	70,000
Operating income (loss)	\$(20,000)

What is Division X's segment margin?

- ☐ a. \$(20,000)
- ☐ b. \$10,000
- ☒ c. \$20,000
- ☐ d. \$(10,000)



Feedback

Check My Work
Correct
Post-Submission

SUPPORTING CALCULATIONS:
\$50,000 – \$30,000 = \$20,000

23. MC.08.92

Nolan Company has the following information pertaining to its two divisions for last year:

	Division X	Division Y
Variable selling and admin. expenses	\$70,000	\$90,000
Direct fixed overhead	35,000	100,000
Sales	200,000	400,000
Direct fixed selling and admin. expenses	30,000	70,000
Variable cost of goods sold	40,000	100,000

Common expenses are \$25,000 for the year.

What is the income for Nolan Company?

- ☐ a. \$300,000
☐ b. \$325,000
☒ c. \$40,000
☐ d. \$65,000

Feedback

Check My Work

Correct

Post-Submission

SUPPORTING CALCULATIONS:

Segment income for Y: $\$400,000 - \$90,000 - \$100,000 - \$70,000 - \$100,000 = \$40,000$

Segment income for X: $\$200,000 - \$70,000 - \$35,000 - \$30,000 - \$40,000 = \$25,000$

Income for Nauman Company = $\$25,000 + \$40,000 - \$25,000 = \$40,000$

24. MC.08.120

Wohlers Company produces two types of products, X and Y, with unit contribution margins of \$25 and \$10 respectively. Each product must be cut by a special machine. The company owns eight machines that together provide 40,000 hours of machine time per year. Product X requires 2 hours of machine time, and Product Y requires 0.5 hours of machine time. There are no other constraints. What is the contribution margin per hour of machine time for Product X?

- ☒ a. \$12.50
☐ b. \$10.00
☐ c. \$30.00
☐ d. \$25.00

Feedback

Check My Work

Correct

Post-Submission

Contribution margin per unit \$25.00

Required machine time per unit $\div 2$

Contribution margin per hour of machine time \$12.50

25. MC.08.124

Wohlers Company produces two types of products, X and Y, with unit contribution margins of \$25 and \$10 respectively. Each product must be cut by a special machine. The company owns eight machines that together provide 40,000 hours of machine time per year. Product X requires 2 hours of machine time, and Product Y requires 0.5 hours of machine time. There are no other constraints. What is the total contribution margin for the optimal mix?

- ☒ a. \$800,000
☐ b. \$200,000
☐ c. \$400,000
☐ d. \$600,000

Feedback

Check My Work

Correct

Post-Submission

	Product X	Product Y
Contribution margin per unit	\$25.00	\$10.00
Required machine time per unit	$\div 2$	$\div 0.5$
Contribution margin per hour of machine time	<u><u>\$12.50</u></u>	<u><u>\$20.00</u></u>

Since Product Y yields \$20 of contribution margin per hour of machine time, all machine time should be devoted to the production of Product Y.
Units of Product Y = 40,000 total hours / 0.5 hours per Product Y = 80,000 units

The optimal mix is Product Y = 80,000 units and Product X = 0.
Contribution margin = 80,000 × \$10.00 = \$800,000.

26. MC.08.131

Stadium Company charges cost plus 60%. If the price of an item is \$260, what is the item's cost?



- ☐ a. \$125.50
- ☐ b. \$180
- ☒ c. \$162.50
- ☐ d. \$100

Feedback

Check My Work
Correct
Post-Submission

Price = COGS + (Markup × COGS) or Price = COGS × (1 + Markup) thus COGS = Price ÷ (1 + Markup)

COGS = \$260 ÷ (1 + .60)

COGS = \$260 ÷ 1.60

COGS = \$162.50

27. MC.08.132

Mattson Construction charges each customer a price equal to the cost of direct materials, direct labor, and overhead plus 40%. Job 1845 included the following costs:

Direct materials	\$39,000
Direct labor	67,000
Overhead	26,000

What price is charged for Job 1845?



- ☐ a. \$86,000
- ☒ b. \$184,800
- ☐ c. \$42,400
- ☐ d. \$106,000

Feedback

Check My Work
Correct
Post-Submission

Price = (\$39,000 + \$67,000 + \$26,000) × 1.40 = \$184,800

28. MC.08.136

Hyacinth Company is designing an all-in-one grill and cooler aimed at sports fans. The company believes that the product can be sold for \$250. Hyacinth requires a 20% profit on new products. What is the target cost of the all-in-one grill and cooler?



- ☐ a. \$122
- ☒ b. \$200
- ☐ c. \$251
- ☐ d. \$145

Check My Work
Correct
Post-Submission

$$\text{Target Cost} = \text{Target Price} - \text{Desired Profit}^* = \$250 - \$50 = \$200$$

$$^*\text{Desired Profit} = \text{Profit Margin} \times \text{Selling Price} = 0.20 \times \$250 = \$50$$

29. MC.08.139

Irisen Company has designed a caller ID machine with a large screen that can be seen easily from across the room. The sales department believes that this product can be sold for \$40 each. Irisen requires that all new products yield 10% profit. What is the target cost of the new product?



- ☐ a. \$28
☐ b. \$41
☒ c. \$36
☐ d. \$14

Check My Work
Correct
Post-Submission

$$\text{Target Cost} = \text{Target Price} - \text{Desired Profit}^* = \$40 - \$4 = \$36$$

$$^*\text{Desired Profit} = \text{Profit Margin} \times \text{Selling Price} = 0.10 \times \$40 = \$4$$

30. PR.08.186.ALGO (Algorithmic)

Robinson Corporation manufactures a single product with the following unit costs for 3,500 units:

Direct materials	\$50
Direct labor	25
Factory overhead (40% variable)	100
Selling expenses (60% variable)	50
Administrative expenses (20% variable)	15
Total per unit	<u>\$240</u>

Recently, a company approached Robinson Corporation about buying 1,000 units for \$265. Currently, the models are sold to dealers for \$466.50. Robinson's capacity is sufficient to produce the extra 1,000 units. No additional selling expenses would be incurred on the special order.

Required:

A. What is the profit earned by Robinson Corporation on the original 3,500 units?

\$

792750

B. Should Robinson accept the special order if its goal is to maximize short-run profits?

Yes

What will be the effect on income?

\$

147000

C. Determine the minimum price Robinson would want to receive in order to increase profits by \$9,500 on the special order. **Round your answer to the nearest cent.**

\$

127.50

per unit

Check My Work
Correct
Post-Submission

A.

Sales ($3,500 \times \$466.5$)	\$1,632,750
Less: Costs ($3,500 \times \$240$)	<u>840,000</u>
Net income	<u><u>\$792,750</u></u>

B. Yes, profit will increase by \$147,000:

Increase in sales ($1,000 \times \$265$)	\$ 265,000
Less:	
Increase in direct materials ($1,000 \times \$50$)	(50,000)
Increase in direct labor ($1,000 \times \$25$)	(25,000)
Increase in var. overhead ($1,000 \times \$100 \times 0.40$)	(40,000)
Increase in var. adm. ($1,000 \times \$15 \times 0.20$)	<u>(3,000)</u>
Increase in profits	<u><u>\$147,000</u></u>

C. $\$50 + \$25 + (\$100 \times 0.40) + (\$15 \times 0.20) + (\$9,500 \div 1,000) = \127.50 per unit

Solution

Robinson Corporation manufactures a single product with the following unit costs for 3,500 units:

Direct materials	\$50
Direct labor	25
Factory overhead (40% variable)	100
Selling expenses (60% variable)	50
Administrative expenses (20% variable)	<u>15</u>
Total per unit	<u><u>\$240</u></u>

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Required:

A. What is the profit earned by Robinson Corporation on the original 3,500 units?

\$

B. Should Robinson accept the special order if its goal is to maximize short-run profits?

What will be the effect on income?

\$

C. Determine the minimum price Robinson would want to receive in order to increase profits by \$9,500 on the special order. **Round your answer to the nearest cent.**

\$ per unit

31. PR.08.189.ALGO (Algorithmic)

Greenland Company prints a variety of publications and colored inserts for promotional brochures. Currently, Greenland produces its own ink, including a special metallic color. India Inks has offered to supply Greenland with the 28,000 ounces of metallic ink that it needs each year for \$1.44 per ounce. Greenland is interested because this is a particularly difficult ink to make. The purchasing department must make special efforts to locate suppliers; the

metallic component requires special handling, and since the metallic ink uses machinery that is also used to make other colors of ink, the machinery must be cleaned very well before every batch of metallic ink. The accounting department supplied the following unit costs:

Direct materials	\$0.31
Direct labor	0.13
Variable overhead	0.01
Fixed overhead*	0.57

*Fixed overhead is applied on the basis of a plantwide rate based on direct labor hours.

Upon hearing of the analysis of the cost of making the metallic ink in-house versus buying it from an outside supplier, Jim Webb, the production supervisor, said, "That's nuts! This ink is a real pain to make and \$1.44 per ounce sounds like a bargain to me!" Based on Jim's feelings, Anna Ruiz (a new CMA in the accounting office) conducted an ABC analysis of ink production. She came up with the same direct materials, direct labor, and variable overhead, as well as the following information on activities required by metallic ink production:

Setups	\$69,500	400 setups per year
Purchasing	\$205,000	8,600 purchase orders per year

The metallic ink requires 300 purchase orders per year and 40 setups.

In calculations, round values to the cent and use in subsequent computations. For operating income, round to nearest whole dollar.

A. If Greenland purchases the ink from the outside supplier, operating income would be \$ 
13440
higher   .

B. What is the highest price per ounce that Greenland would pay the outside company for the ink? **Round your answer to the nearest cent.**
\$ 
0.96

Feedback

Check My Work
Correct
Post-Submission

A. \$13,440 higher
 $[\$1.44 - (\$0.31 + \$0.13 + \$0.01 + \$0.51^*)] \times 28,000 = \$13,440$
*\$0.51 (\$0.25 + \$0.26) is the per-ounce cost of setups and purchasing
 $\text{Setups} = [(\$69,500 \div 400) \times 40] \div 28,000 = \0.25
 $\text{Purchasing} = [(\$205,000 \div 8,600) \times 300] \div 28,000 = \0.26

B. Since \$0.96 (\$0.31 + \$0.13 + \$0.01 + \$0.51) is the avoidable cost of making the ink in-house, Greenland would not pay an outside company more than this.

Solution

Greenland Company prints a variety of publications and colored inserts for promotional brochures. Currently, Greenland produces its own ink, including a special metallic color. India Inks has offered to supply Greenland with the 28,000 ounces of metallic ink that it needs each year for \$1.44 per ounce. Greenland is interested because this is a particularly difficult ink to make. The purchasing department must make special efforts to locate suppliers; the metallic component requires special handling, and since the metallic ink uses machinery that is also used to make other colors of ink, the machinery must be cleaned very well before every batch of metallic ink. The accounting department supplied the following unit costs:

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Setups \$69,500 400 setups per year
Purchasing \$205,000 8,600 purchase orders per year

The metallic ink requires 300 purchase orders per year and 40 setups.

In calculations, round values to the cent and use in subsequent computations. For operating income, round to nearest whole dollar.

A. If Greenland purchases the ink from the outside supplier, operating income would be \$.

B. What is the highest price per ounce that Greenland would pay the outside company for the ink? **Round your answer to the nearest cent.**
\$

32. PR.08.190.ALGO (Algorithmic)

Sherpa Company manufactures tents and sleeping bags. Tents are priced at \$84 and have variable cost of \$60 and direct fixed costs of \$125,000. Sleeping bags are priced at \$62 and have variable cost of \$30 and direct fixed costs of \$68,000. Common fixed costs equal \$225,000. Last year, Sherpa sold 5,300 tents and 9,000 sleeping bags.

Use the minus sign to indicate negative numbers.

A. What was the segment margin for tents last year?
\$ 
2200

B. What was the segment margin for sleeping bags last year?
\$ 
220000

C. What was Sherpa's operating income (loss) last year?
\$ 
-2800

D. If Sherpa stopped making tents, what would operating income (loss) be?
\$ 
-5000

Feedback

Check My Work

Correct

Post-Submission

	Sleeping		
	Tents	Bags	Total
Sales ($\$84 \times 5,300$; $\$62 \times 9,000$)	\$445,200	\$558,000	\$1,003,200
Less: Variable cost ($\$60 \times 5,300$; $\$30 \times 9,000$)	318,000	270,000	588,000
Contribution margin	\$127,200	\$288,000	\$415,200
Less: Direct fixed overhead	125,000	68,000	193,000
Segment margin	<u>\$ 2,200</u>	<u>\$ 220,000</u>	\$ 222,200
Less: Common fixed overhead			225,000
Operating income			<u><u>\$ (2,800)</u></u>

A. Segment margin for tents = \$2,200

B. Segment margin for sleeping bags = \$220,000

C. Operating income = \$(2,800)

D. Operating income without tents = \$(5,000) (\$220,000 – \$225,000)

Solution

Sherpa Company manufactures tents and sleeping bags. Tents are priced at \$84 and have variable cost of \$60 and direct fixed costs of \$125,000. Sleeping bags are priced at \$62 and have variable cost of \$30 and direct fixed costs of \$68,000. Common fixed costs equal \$225,000. Last year, Sherpa sold 5,300 tents and 9,000 sleeping bags.

Use the minus sign to indicate negative numbers.

A. What was the segment margin for tents last year?

\$ 2200

B. What was the segment margin for sleeping bags last year?

\$ 220000

C. What was Sherpa's operating income (loss) last year?

\$ -2800

D. If Sherpa stopped making tents, what would operating income (loss) be?

\$ -5000

33. PR.08.192.ALGO (Algorithmic)

Mickey Company manufactures three joint products: X, Y, and Z. The cost of the joint process is \$33,000. Information about the three products follows:

	X	Y	Z
Anticipated production	5,500 lbs.	10,800 lbs.	2,200 lbs.
Selling price/lb. at split-off	\$2.00	\$1.00	\$3.25
Additional processing costs/lb. after split-off			
(all variable)	\$0.50	\$1.50	\$1.25
Selling price/lb. after further processing	\$2.00	\$4.50	\$5.75
Allocated joint costs	\$11,900	\$10,700	\$6,000

A. Determine whether each product should be sold at split-off or processed further.

Product	Decision
X	Split-off ✓
Y	Process further ✓
Z	Process further ✓

B. Determine the firm's income if the firm processed all three products beyond split-off.

\$ ✓

17550

Feedback

Check My Work

Correct

Post-Submission

A.

Sell at	Process	Decision
Split-Off	Further	

Then Sell

X	\$11,000	\$11,000
		(2,750) Sell at split-off
		<u>\$8,250</u>
Y	\$10,800	\$48,600
		(16,200) Process further
		<u>\$32,400</u>
Z	\$7,150	\$12,650
		(2,750) Process further
		<u>\$9,900</u>

The joint costs are not relevant to the decision.

B. \$17,550 (\$9,900 + \$32,400 + \$8,250 – \$33,000)

Solution

Mickey Company manufactures three joint products: X, Y, and Z. The cost of the joint process is \$33,000. Information about the three products follows:

	<u>X</u>	<u>Y</u>	<u>Z</u>
Anticipated production	5,500 lbs.	10,800 lbs.	2,200 lbs.
Selling price/lb. at split-off	\$2.00	\$1.00	\$3.25
Additional processing costs/lb. after split-off (all variable)	\$0.50	\$1.50	\$1.25
Selling price/lb. after further processing	\$2.00	\$4.50	\$5.75
Allocated joint costs	\$11,900	\$10,700	\$6,000

A. Determine whether each product should be sold at split-off or processed further.

Product	Decision
X	Split-off
Y	Process further
Z	Process further

B. Determine the firm's income if the firm processed all three products beyond split-off.

\$ 17550

34. PR.08.193.ALGO (Algorithmic)

Information pertaining to Mario Co. is as follows:

Leather Jackets Suede Jackets

Sales	\$473,000	\$600,000
Variable cost of goods sold	141,000	204,500
Direct fixed overhead	20,500	37,500

A sales commission of 3% of sales is paid for each of the two product lines. Direct fixed selling and administrative expense is estimated to be \$33,000 for the leather jackets and \$68,100 for the suede jackets. Common fixed overhead for the factory is estimated to be \$87,000 and common selling and administrative expense is estimated to be \$16,800.

Required:

Prepare a segmented income statement for Mario Co. for the coming year, using variable costing.

Mario Co. **Segmented Income Statement**

For the Coming Year

	Leather Jackets	Suede Jackets	Total
	\$ ✓	\$ ✓	\$ ✓
Sales	473000	600000	1073000
	✓	✓	✓
Variable cost of goods sold	141000	204500	345500
	✓	✓	✓
Variable selling expense	14190	18000	32190
	✓	✓	✓
Contribution margin	317810	377500	695310
Less direct fixed expenses:			
	✓	✓	✓
Direct fixed overhead	20500	37500	58000
	✓	✓	✓
Direct selling and administrative	33000	68100	101100
	✓	✓	✓
Segment margin	264310	271900	536210
Less common fixed expenses:			
			✓
Common fixed overhead			87000
			✓
Common selling and administrative			16800
			✓
Operating income			432410

Feedback

Check My Work
Correct
Solution

Information pertaining to Mario Co. is as follows:

Leather Jackets Suede Jackets

Sales	\$473,000	\$600,000
Variable cost of goods sold	141,000	204,500
Direct fixed overhead	20,500	37,500

A sales commission of 3% of sales is paid for each of the two product lines. Direct fixed selling and administrative expense is estimated to be \$33,000 for the leather jackets and \$68,100 for the suede jackets. Common fixed overhead for the factory is estimated to be \$87,000 and common selling and administrative expense is estimated to be \$16,800.

Required:

Prepare a segmented income statement for Mario Co. for the coming year, using variable costing.

**Mario Co.
Segmented Income Statement
For the Coming Year**

	Leather Jackets	Suede Jackets	Total
Sales	\$ 473000	\$ 600000	\$ 1073000
Variable cost of goods sold	141000	204500	345500
Variable selling expense	14190	18000	32190

Contribution margin	\$ 317810	\$ 377500	\$ 695310
Less direct fixed expenses:			
Direct fixed overhead	20500	37500	58000
Direct selling and administrative	33000	68100	101100
Segment margin	\$ 264310	\$ 271900	\$ 536210
Less common fixed expenses:			
Common fixed overhead			87000
Common selling and administrative			16800
Operating income			\$ 432410

35. PR.08.195.ALGO (Algorithmic)

The operations of Grant Corporation are divided into the Fix Division and the Split Division. Projections for the next year are as follows:

	Fix Division	Split Division	Total
Sales revenue	\$57,600	\$35,000	\$92,600
Variable expenses	22,000	16,600	38,600
Contribution margin	\$35,600	\$18,400	\$54,000
Direct fixed expenses	14,700	35,000	49,700
Segment margin	\$20,900	\$(16,600)	\$4,300
Allocated common costs	9,000	7,700	16,700
Total relevant benefit (loss)	\$11,900	\$(24,300)	\$-12,400

Required:

A. Determine operating income for Grant Corporation as a whole if the Split Division is dropped.

\$

4200

B. Should the Split Division be eliminated?

Yes

Feedback

Check My Work
Correct
Post-Submission

A.

Sales revenue	\$57,600
Variable expenses	22,000
Contribution margin	\$35,600
Direct fixed expenses	14,700
Segment margin	\$20,900
Allocated common costs (\$9,000 + \$7,700)	16,700
Operating income	\$4,200

B. Yes, the Split division should be dropped since it has a negative segment margin of \$16,600. Dropping the Split division increases the firm's income by \$16,600.

Solution

The operations of Grant Corporation are divided into the Fix Division and the Split Division. Projections for the next year are as follows:

	Fix	Split	
	Division	Division	Total
Sales revenue	\$57,600	\$35,000	\$92,600
Variable expenses	22,000	16,600	38,600
Contribution margin	\$35,600	\$18,400	\$54,000
Direct fixed expenses	14,700	35,000	49,700
Segment margin	\$20,900	\$(16,600)	\$4,300
Allocated common costs	9,000	7,700	16,700
Total relevant benefit (loss)	\$11,900	\$(24,300)	\$-12,400

Required:

A. Determine operating income for Grant Corporation as a whole if the Split Division is dropped.

\$

B. Should the Split Division be eliminated?

36. PR.08.197.ALGO (Algorithmic)

Gordon Company produces two types of gears, Gear Q and Gear S, with unit contribution margins of \$3 and \$6, respectively. Each gear must spend time on a special machine. The firm owns 10 machines that together provide 33,000 hours of machine time per year. Gear Q requires 0.07 hours of machine time; Gear S requires 0.48 hours of machine time.

Required:

If an amount is zero, enter "0".

A. What is the contribution margin per hour of machine time for Gear Q and Gear S? **Round your answers to the nearest cent.**

Gear Q ✓

Gear S ✓

B. If Gordon faces only the production constraint (33,000 hours of machine time), how many units of Gear Q and Gear S should be produced? **Round your answers to the nearest whole number.**

Gear Q ✓

Gear S ✓

C. What is the total contribution margin from this product mix? **Round your answer to the nearest dollar.**

\$ ✓

D. Now suppose that Gordon cannot sell more than 180,000 units of each type of gear. How many units of Gear Q and Gear S should be produced? **Round your answers to the nearest whole number.**

Gear Q ✓

Gear S ✓

E. What is the total contribution margin from this product mix? **Round your answer to the nearest dollar.**

\$ ✓

Check My Work
Correct
Post-Submission

A.

	Gear Q	Gear S
Contribution margin per unit	\$3.00	\$6.00
Hours of machine time	÷ 0.07	÷ 0.48
Contribution margin per hour of machine time	<u>\$42.86</u>	<u>\$12.50</u>

B. Since Gear Q has the higher contribution margin per hour of the scarce resource, Gordon should produce only Gear Q, and no Gear S.

$$\text{Gear Q} = 33,000 \div 0.07 = 471,429 \text{ units}$$

$$\text{Gear S} = 0 \text{ units}$$

C. Total Contribution Margin = $\$3.00 \times 471,429 = \$1,414,287$

D. Gear Q = 180,000 units

$$\text{Remaining machine time} = 33,000 \text{ hours} - (180,000 \times 0.07) = 20,400 \text{ machine hours}$$

$$\text{Gear S} = 20,400 \div 0.48 = 42,500 \text{ units}$$

E. Total Contribution Margin = $(\$3.00 \times 180,000) + (\$6.00 \times 42,500) = \$795,000$

Solution

Gordon Company produces two types of gears, Gear Q and Gear S, with unit contribution margins of \$3 and \$6, respectively. Each gear must spend time on a special machine. The firm owns 10 machines that together provide 33,000 hours of machine time per year. Gear Q requires 0.07 hours of machine time; Gear S requires 0.48 hours of machine time.

Required:

If an amount is zero, enter "0".

A. What is the contribution margin per hour of machine time for Gear Q and Gear S? **Round your answers to the nearest cent.**

Gear Q \$

Gear S \$

B. If Gordon faces only the production constraint (33,000 hours of machine time), how many units of Gear Q and Gear S should be produced? **Round your answers to the nearest whole number.**

Gear Q

Gear S

C. What is the total contribution margin from this product mix? **Round your answer to the nearest dollar.**

\$

D. Now suppose that Gordon cannot sell more than 180,000 units of each type of gear. How many units of Gear Q and Gear S should be produced? **Round your answers to the nearest whole number.**

Gear Q

Gear S

E. What is the total contribution margin from this product mix? **Round your answer to the nearest dollar.**

\$

The Exchange Company is in the process of developing a new product called LS500. The company requires a 43% profit. LS500's current design carries with it a total cost of \$108.

Required:

A. What is the sales price of LS500 using markup costing? Round your answer to two decimal places.

\$ 154.44

B. Assume that the Exchange Company's marketing department has determined that consumers are willing to pay \$147 for LS500. What is the target cost for this product? Round your answer to the nearest dollar.

\$ 84

Feedback

Check My Work
Correct
Post-Submission

A. Sales Price = \$108 × 1.43 = \$154.44

B. Desired Profit per Unit = 0.43 × \$147 = \$63
Target Cost = \$147 – \$63 = \$84

Solution

The Exchange Company is in the process of developing a new product called LS500. The company requires a 43% profit. LS500's current design carries with it a total cost of \$108.

Required:

A. What is the sales price of LS500 using markup costing? Round your answer to two decimal places.

\$ 154.44

B. Assume that the Exchange Company's marketing department has determined that consumers are willing to pay \$147 for LS500. What is the target cost for this product? Round your answer to the nearest dollar.

\$ 84

38. PR.08.199.ALGO (Algorithmic)

Auden makes three types of vitamin supplements, all of which require the use of encapsulating machines that have a capacity of 7,000 hours. Information on the three types (per case) is as follows:

	Basic	Vita-Stress	Antioxidant+
Selling price	\$92	\$129	\$158
Variable cost	50	63	81
Machine hours required	0.1	0.5	0.9

If required, round your answers to the nearest cent.

A. What is the contribution margin per case for each type?

Basic \$ 42
Vita-Stress \$ 66
Antioxidant+ \$ 77

B. What is the contribution margin per hour of machine time for each type? Round your answer to two decimal places.

	\$	✓
Basic	420	
	\$	✓
Vita-Stress	132	
	\$	✓
Antioxidant+	85.56	

C. Based on your analysis in requirement B, if Auden can sell all that it can make of all of the products, how many of each type should be sold to maximize total contribution margin? **If an amount is zero enter "0" as your answer.**

	✓
Basic	70000
	cases
	✓
Vita-Stress	0
	cases
	✓
Antioxidant+	0
	cases

Feedback

Check My Work
Correct
Post-Submission

A. and B.

	Basic	Vita-Stress	Antioxidant+
Price	\$92	\$129	\$158
- Variable cost	50	63	81
Contribution margin per unit	\$42	\$66	\$77
Hours of machine time required	÷ 0.1	÷ 0.5	÷ 0.9
Contribution margin per hour of machine time	\$420	\$132	\$85.56

C.

Basic has the highest contribution margin per machine hour. Therefore, Auden should devote all of its encapsulating machine time to the production of Basic and make 70,000 cases of it.

Basic = (1 hour ÷ 0.10 case per hour) × 7,000 hours = 70,000 cases

Vita-Stress = 0 cases

Antioxidant+ = 0 cases

Solution

Auden makes three types of vitamin supplements, all of which require the use of encapsulating machines that have a capacity of 7,000 hours. Information on the three types (per case) is as follows:

	Basic	Vita-Stress	Antioxidant+
Selling price	\$92	\$129	\$158
Variable cost	50	63	81
Machine hours required	0.1	0.5	0.9

If required, round your answers to the nearest cent.

A. What is the contribution margin per case for each type?

Basic	\$	42
Vita-Stress	\$	66

Antioxidant+ \$

B. What is the contribution margin per hour of machine time for each type? **Round your answer to two decimal places.**

Basic \$

Vita-Stress \$

Antioxidant+ \$

C. Based on your analysis in requirement B, if Auden can sell all that it can make of all of the products, how many of each type should be sold to maximize total contribution margin? **If an amount is zero enter "0" as your answer.**

Basic cases

Vita-Stress cases

Antioxidant+ cases